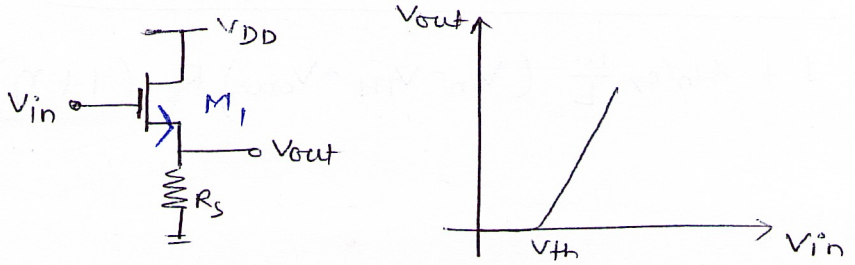
Source follower :-

→ Since, to achieve high Vtg gain with limited supply Vtg, the load impedence must be as large as possible. (C-S stage)

→ But, If such a stage is to drive low impedence load, the buffer must be place after the amplifier so as to drive the load ~~with load~~ with negligible loss of signal level. The source follower also called Common-Drain (C-C in BJT) stage can opesate as a voltage buffer.



(a) Source follower (b) its i/p - o/p characteristics.

→ the source follower senses the signal at the gate and drives the load at the source potential to follow the gate voltage.

Calculation of gain :-

if $V_{in} < V_{TH}$, ⇒ M_1 off ⇒ $V_{out} = 0$

$V_{in} > V_{TH}$ ⇒ M_1 turns on in Saturation ⇒ I_{D_1} flows

∴ i/p - o/p char,

$$\underbrace{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})^2}_{I_{D_1}} R_S = V_{out}$$

Small gain of2 circuit by differentiating both side w.r.t. V_{in}

$$\therefore \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\,2(V_{in}-V_{TH}-V_{out})\left(1-\frac{\partial V_{TH}}{\partial V_{in}}-\frac{\partial V_{out}}{\partial V_{in}}\right)R_S$$

$$= \frac{\partial V_{out}}{\partial V_{in}}$$

Chain rule

$$\therefore \frac{\partial V_{TH}}{\partial V_{in}} = \eta\frac{\partial V_{out}}{\partial V_{in}}$$

$$\left\{\because \frac{\partial V_{TH}}{\partial V_{in}} = \frac{\partial V_{TH}}{\partial V_{out}}\cdot\frac{\partial V_{out}}{\partial V_{in}} = \eta\cdot\frac{\partial V_{out}}{\partial V_{in}}\right\}$$

$$\frac{1}{2}\mu_n C_{ox}\frac{W}{L}\,2(V_{in}-V_{TH}-V_{out})\left(1-\eta\cdot\frac{\partial V_{out}}{\partial V_{in}}-\frac{\partial V_{out}}{\partial V_{in}}\right)R_S$$

$$=\frac{\partial V_{out}}{\partial V_{in}}$$

$$\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})R_S-\frac{\partial V_{out}}{\partial V_{in}}(\eta+1)\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})R_S$$

$$=\frac{\partial V_{out}}{\partial V_{in}}$$

$$\frac{\partial V_{out}}{\partial V_{in}}=\frac{\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})R_S}{1+\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})R_S(1+\eta)} \quad (34)$$

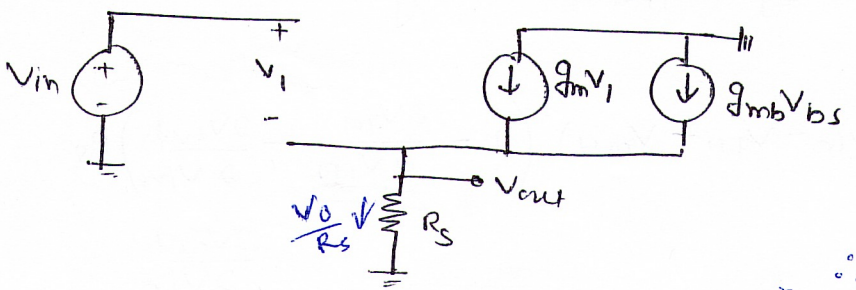
$$g_m=\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})$$

$$A_V=\frac{g_m R_S}{1+\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{TH}-V_{out})R_S(1+\eta)}$$

$$\boxed{A_v=\frac{g_m R_S}{1+(g_m+g_{mb})R_S}} \quad (35)$$

Note:-

Above result can be easily obtained from following small signal equivalent circuit.



from above Ckt, $V_{in}-V_1=V_{out}$

$V_{bs}=-V_{out}$

$\therefore g_m V_1+g_{mb}(-V_{out})=\frac{V_{out}}{R_S}$

$\therefore \frac{V_{out}}{V_1}$

$$\therefore \quad g_m(V_{in} - V_{out}) - g_{mb}V_{out} = \frac{V_{out}}{R_S}$$

$$\boxed{\frac{V_{out}}{V_{in}} = \frac{g_m R_S}{[1 + (g_m + g_{mb})R_S]}} \quad (<1)$$

, In phase output

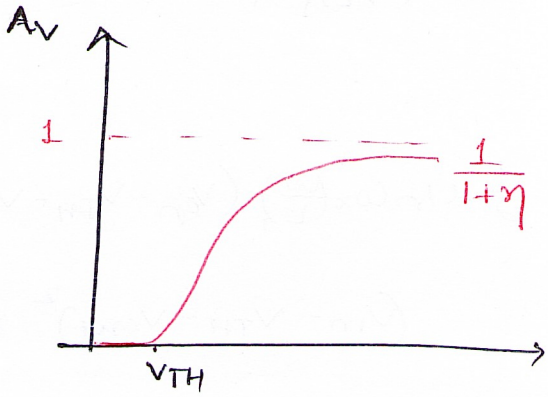


Fig: Voltage gain of source follower versus I/p voltage.

as drain current ↑es & g_m ↑es $\Rightarrow$ $A_V \simeq \frac{g_m}{(g_m + g_{mb})} = \left(\frac{1}{1+\eta}\right)$

$\because$ η slowly ↓es with V_{out} $\Rightarrow$ $A_V \simeq 1$, but for typical allowable source-bulk voltages, η remains greater than roughly 0.2

Issues:-

now in above case, the drain current of M1 heavily depends on the I/p dc level. for exp. if Vin changes from 1.5V to 2V, I_D may ↑es by factor of 2 and hence $V_{GS} - V_{TH}$ by $\sqrt{2}$

$\Rightarrow$ Introducing non linearity; non-linearity in the I/p & O/p characteristics.

Solution:-

So to overcome above issue resister can be replace by a current source. The current source itself implemented as an NMOS transistor operating in the saturation region.

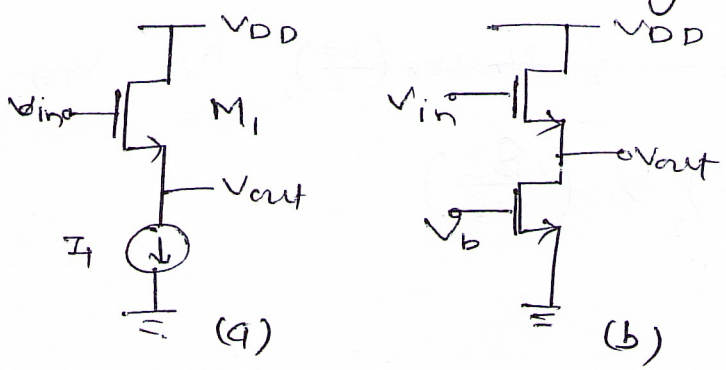


Fig:- Source follower using an NMOS transistor as current source

exp:- Suppose in above fig. (a) $\left(\frac{W}{L}\right)_1 = 20/0.5$, $I_1 = 200\,\mu A$

$V_{THO} = 0.6V$, $2\phi_F = 0.7\,V$, $\mu_n C_{ox} = 50\,\mu A/V^2$ & $\gamma = 0.4$

(a) calculate V_{out} for $V_{in} = 1.2V$

(b) If I_1 is implemented as M_2 in fig (b), find the min. value of $(W/L)_2$ for which M_2 remains in saturated.

(a)

$\because$ M_1 in saturation,

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{in} - V_{TH} - V_{out})^2 = I_D$$

$$(V_{in} - V_{TH} - V_{out})^2 = \frac{2 I_D}{\mu_n C_{ox} (W/L)_1}$$

but us assume $V_{TH} \simeq 0.6$, $\therefore$ $V_{out} = 0.153V$

but (*) $V_{TH} = V_{THO} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right)$

$= \simeq 0.635$

$\therefore$ $V_{out} \simeq 0.119V$ (actual)

(b)

V_{DS} of $M_2 \Rightarrow 0.119V$,

M_2 will be saturation only, if $0.119 \geq (V_{GS} - V_{TH})_2$

& $I_D = 200\,\mu A$

$\therefore$ $I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_2 (V_{GS} - V_{TH} - V_{out})^2$

$$\left(\frac{W}{L}\right)_2 \geq \left(\frac{283}{0.5}\right)$$

To gain a better understanding of source followers,

Calculation of o/p resistance:- (for source followers)

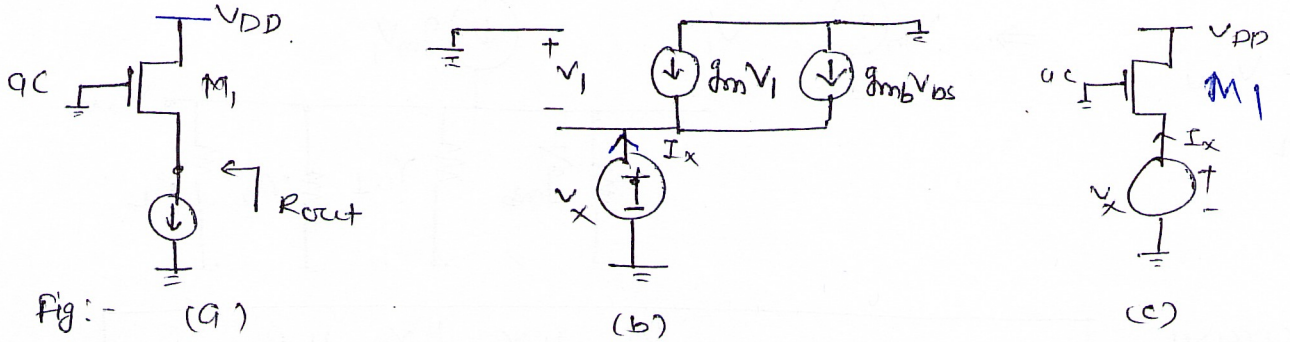


Fig:- (a) (b) (c)

from above ckt,

$$V_1 = -V_x$$

$$\rightarrow \quad I_x - g_m V_x - g_{mb} V_x = 0 \quad \Rightarrow$$

$$R_{out} = \frac{V_x}{I_x} = \frac{1}{g_m + g_{mb}}$$ (3)

(o/p Impedance low)

Conclusion:- body effect ↓es o/p resistance of source follower

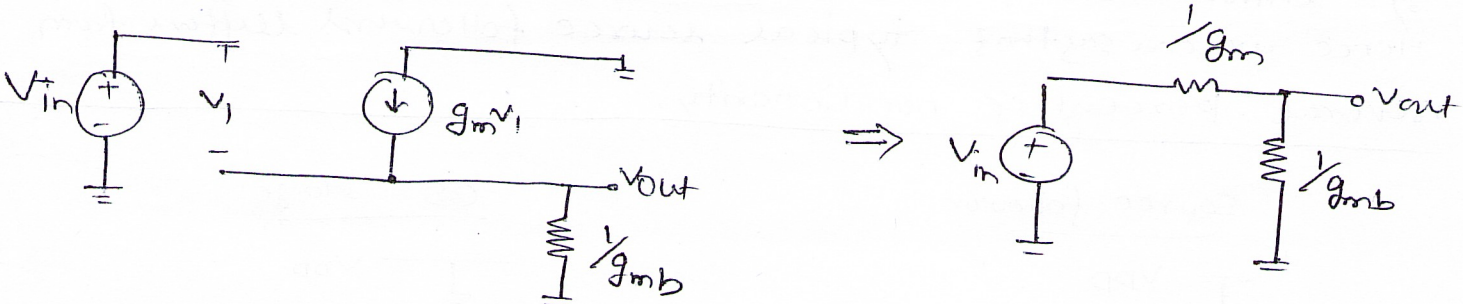
Note:

$$R_{out} = \frac{1}{g_m} \,||\, \frac{1}{g_{mb}} = \frac{1}{g_m + g_{mb}}$$

(can be calculate from small signal model)

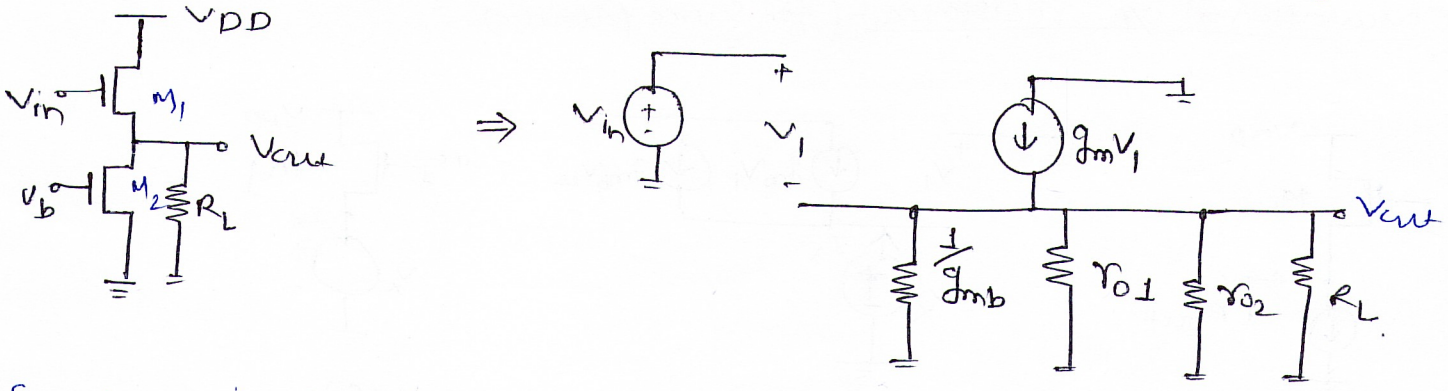
***

Source follower by a thevenins equivalent:-



$$\therefore \quad A_V = \frac{V_{out}}{V_{in}} = \frac{1/g_{mb}}{1/g_m + 1/g_{mb}} = \frac{g_m}{g_m + g_{mb}}$$

Complete practical source follower with finite channel length Model



for above circuit,

$$A_V = \frac{V_{out}}{V_{in}} = \frac{1/g_{mb} \,||\, r_{o1} \,||\, r_{o2} \,||\, R_L}{\left(1/g_{mb} \,||\, r_{o1} \,||\, r_{o2} \,||\, R_L\right) + 1/g_m}$$

(37)

Conclusion:-

1) Source follower exhibit a high i/p impedence and moderate o/p impedence but two drawbacks
   1) non-linearity 2) voltage headroom limitation

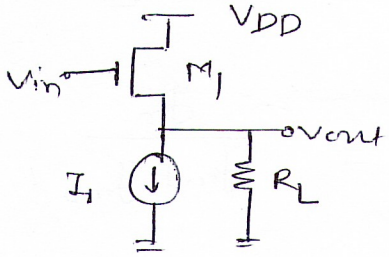
Even

2) if a source follower is biased by an ideal current source, its i/p-o/p char. display some non-linearity due to non-linearity dependence of V_{TH} upon the source potential

3) introduce substantial noise, it is ill-suited to low noise application.

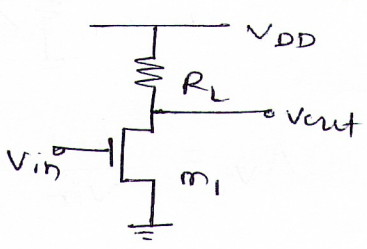
Hence we can say that, typical source follower suffers from several percent of non-linearity.

Source follower



$$A_V = \left.\frac{V_{out}}{V_{in}}\right|_{SF} = \frac{R_L}{R_L + 1/g_{m_1}}$$

CS stage



$$A_V = \left.\frac{V_{out}}{V_{in}}\right|_{CS} = -g_{m_1} R_L$$

3) Common-Gate Stage:-

→ A common-Gate stage senses the i/p at the source and produces o/p at drain.

→ The gate is connected to a dc vtg to establish proper operating conditions.

→ The bias current of M_1 flows through the i/p signal source. Alternatively, M_1 can be biased by a constant current source, with the signal capacitively coupled to the circuit.

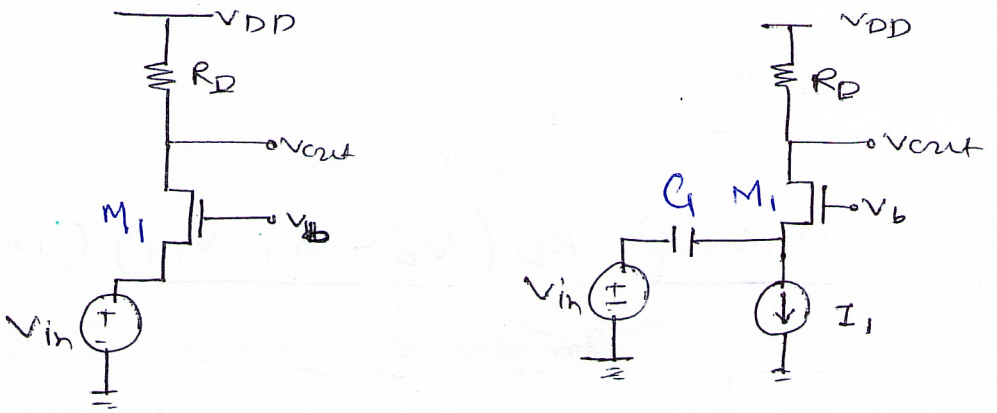


Fig:- (a) (b)

If:- large-signal behaviour,

if $V_{in} \geq V_b - V_{TH}$ M_1 - OFF $\Rightarrow V_{out} = V_{DD}$

for lower V_{in},

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)\left(V_b - V_{in} - V_{TH}\right)^2$$

if M_1 is in saturation,

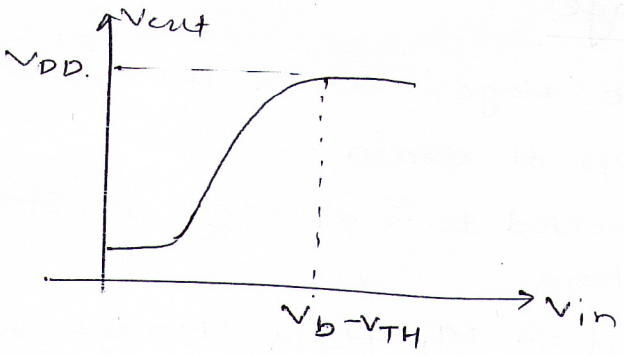
As V_{in} ↓es, so does V_{out}, eventualy driving M_1 driving into a triode region if,

$$V_{DD} - \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)\left(V_b - V_{in} - V_{th}\right)^2 R_D = V_b - V_{TH}$$

if M_1 is in saturation,

$$V_{out} = V_{DD} - \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(V_b - V_{in} - V_{TH}\right)^2 R_D$$

I/p-O/p char:- The input-output characteristic is shown in Fig.



Small signal gain

$$\frac{\partial V_{out}}{\partial V_{in}} = -\mu_n C_{ox} \frac{W}{L} (V_b - V_{in} - V_{TH}) \left(-1 - \frac{\partial V_{TH}}{\partial V_{in}}\right) R_D$$

$$\because \frac{\partial V_{TH}}{\partial V_{in}} = \frac{\partial V_{TH}}{\partial V_{SB}} = \eta$$

$$\therefore \frac{\partial V_{out}}{\partial V_{in}} = \underbrace{\mu_n C_{ox} \frac{W}{L}}_{g_m} R_D \underbrace{(V_b - V_{in} - V_{TH})}_{} (1+\eta)$$

$$A_v = \underbrace{\mu_n C_{ox} \frac{W}{L} (V_b - V_{in} - V_{TH})}_{g_m} \cdot R_D (1+\eta)$$

$$\therefore \boxed{A_v = g_m (1+\eta) R_D} \quad \text{+ive} \quad \text{---(38)}$$

→ body effect ↑es the equivalent transconductance of the stage.

Input impedence

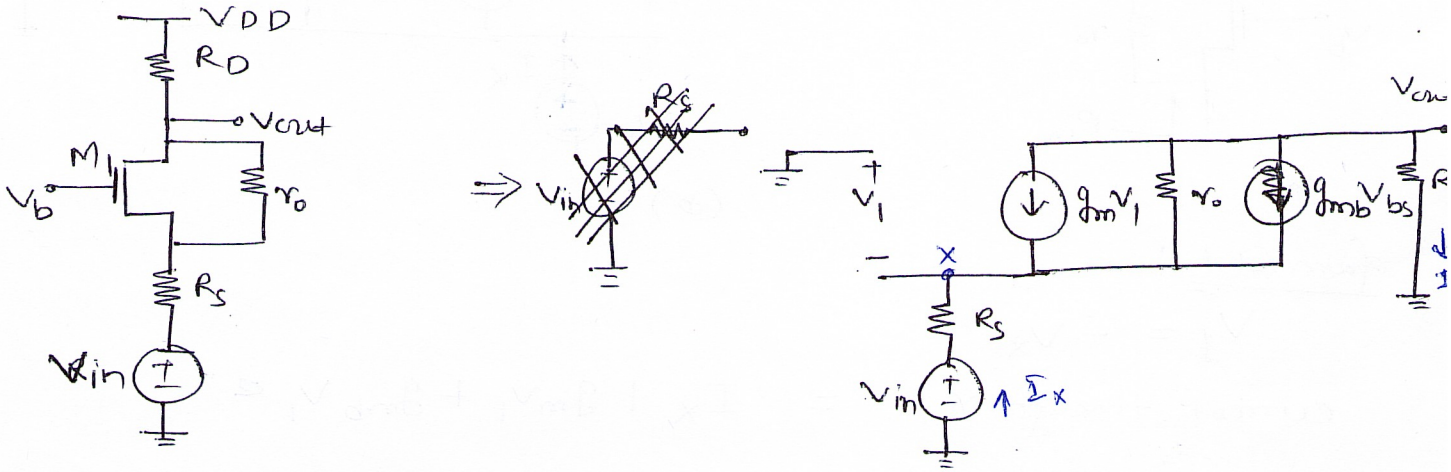
for $\lambda = 0$, the impedence seen at source of M_1,

$$\boxed{R_{in} = \frac{1}{(g_m + g_{mb})} = \frac{1}{g_m (1+\eta)}} \quad (39)$$

{ As we have calculated in eqn (36) }

⇒ body effect decreases the input impedence of common-gate stage. (Low i/p impedence)

CG Stage, taking into account both the ~~arrangement~~ o/p impedence of transister and impedence of the signal source —



from above small signal ckt,

current flowing through R_S, $I_x = \frac{-V_{out}}{R_D}$,

we have,

$$V_1 - \frac{V_{out}}{R_D} R_S + V_{in} = 0 \qquad -(40)$$

current through r_o,

$$-\frac{V_{out}}{R_D} - g_m V_1 - g_{mb} V_1 \text{, we can write,}$$

$$r_o\left(-\frac{V_{out}}{R_D} - g_m V_1 - g_{mb} V_1\right) - \frac{V_{out}}{R_D} \cdot R_S + V_{in} = V_{out} \qquad (41)$$

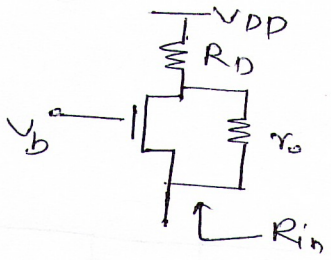
Substitute V_1 from (40) to (41)

$$r_o\left[-\frac{V_{out}}{R_D} - (g_m + g_{mb})\left(\frac{V_{out} R_S}{R_D} - V_{in}\right)\right] - \frac{V_{out}}{R_D} R_S + V_{in} = V_{out}$$

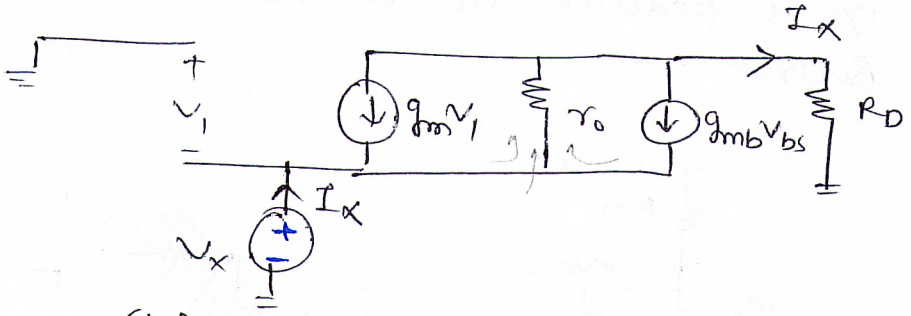
$$\therefore A_V = \frac{V_{out}}{V_{in}} = \left[\frac{(g_m + g_{mb}) r_o + 1}{r_o + (g_m + g_{mb}) r_o R_S + R_S + R_D}\right] R_D$$

⇒ The gain of C-G stage is sightly higher due to body effect compare to CS stage (eqn (32))

I/p & O/p impedance of CG



(a)



(b)

from Ckt :-

$$V_1 = -V_x$$

current through r_o, $= I_x + g_m V_1 + g_{mb} V_1$

$$= I_x - (g_m + g_{mb}) V_x$$

now, voltage across r_o & R_D add up, $= V_x$

$$R_D I_x + r_o (I_x - (g_m + g_{mb}) V_x) = V_x$$

$$\therefore \frac{V_x}{I_x} = \frac{R_D + r_o}{1 + (g_m + g_{mb}) r_o}$$

$$= \frac{R_D}{(g_m + g_{mb}) r_o} + \frac{1}{g_m + g_{mb}} \quad \text{if } (g_m + g_{mb}) r_o \gg 1$$

This result reveals that the drain impedance is divided by $(g_m + g_{mb}) r_o$ when seen at the source.

~~if $(g_m + g_{mb}) \gg 1$~~

if $R_D = 0$

$$\frac{V_x}{I_x} = \frac{r_o}{1 + (g_m + g_{mb}) r_o}$$

$$\frac{V_x}{I_x} = \frac{1}{\left(\frac{1}{r_o} + g_m + g_{mb}\right)}$$

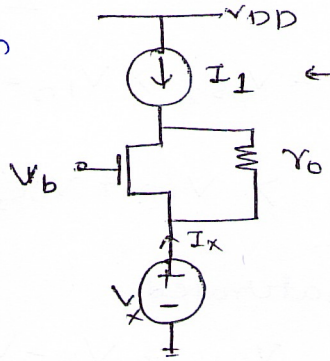
this impedance is same as impedance seen at the source of a source follower.

I/p impedance very low.

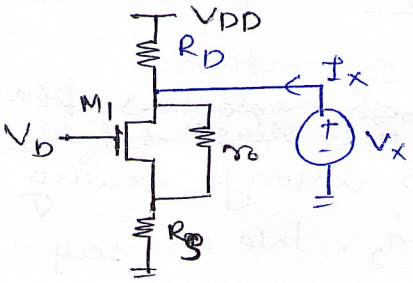
if R_D is replaced with ideal current source, then i/p impedence approches infinity.

⇒ Since the total current through the transister is fixed and equal to I_1, a change in source potential can not change the device current, and hence $I_x = 0$.

→ the i/p impedence of a common gate stage is relatively low only if the load impedence connected to the drain is small.



output impedence of CG stage:- In order to calculate the o/p impedance of the common-gate stage;



$$\frac{V_x}{I_x} = R_{out} = R_D \,||\, \{[1 + (g_m + g_{mb}) r_o] R_S + r_o\}$$

Cascode Stage:- (CS-CG)

→ The cascade of a CS ($CE)_{BJT}$ stage and a CG ($CB)_{BJT}$ state is called cascode topology.

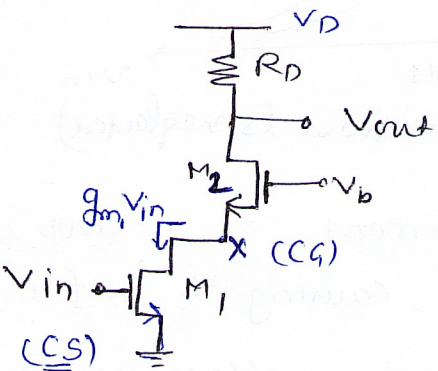


Fig-1

→ M_1 generates a small-signal drain current proportional to Vin and M_2 simply routes the current to R_D.

→ M_1 & M_2 carries equal current.

→ There are many advantages of the cascode topology over a simple CS stage.

For, M_1 to operate in saturation,

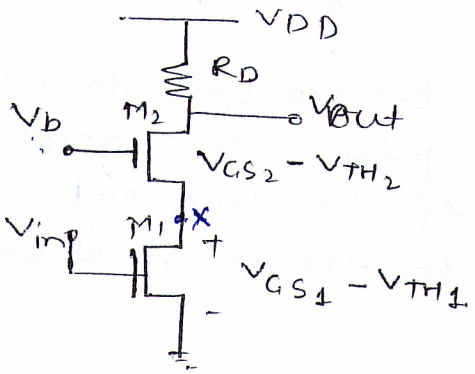
$$V_x > V_{in} - V_{TH_1}$$

if M_1 & M_2 both are in saturation, then V_x is determined by V_b.

$$V_x = V_b - V_{GS_2}$$

$$\therefore V_b - V_{GS_2} > V_{in} - V_{TH_1}$$

$$\therefore V_b > V_{in} + V_{GS_2} - V_{TH_1}$$



for M_2 saturated,

$$V_{out} \geq V_b - V_{TH_2}$$

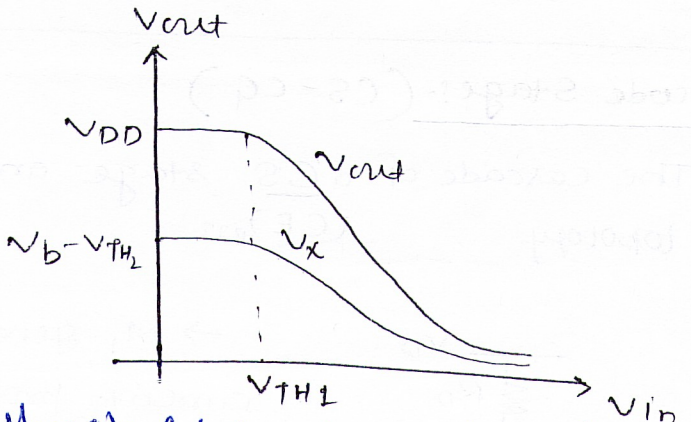
i.e. $V_{out} \geq V_{in} - V_{TH_1} + V_{GS_2} - V_{TH_2}$ if V_b is choosen to place M_1 at the edge of saturation.

Consequently, the minimum o/p level for which both transistors operate in saturation is equal to the overdrive voltage of M_1 & plus that of M_2.

→ Addition of M_2 to the ckt reduces the o/p voltage swing by atleast the overdrive voltage of M_2. We also say M_2 is stacked on top of M_1.

(moving towards)

I/P - o/p char. of cascode :-



Large signal behavior; (Fig. 1)

$V_{in} < V_{TH_1}$ M_1 & M_2 off

$\therefore V_{out} = V_{DD}$

& $V_x \geq V_b - V_{th_2}$ (if subthreshold condition is neglected)

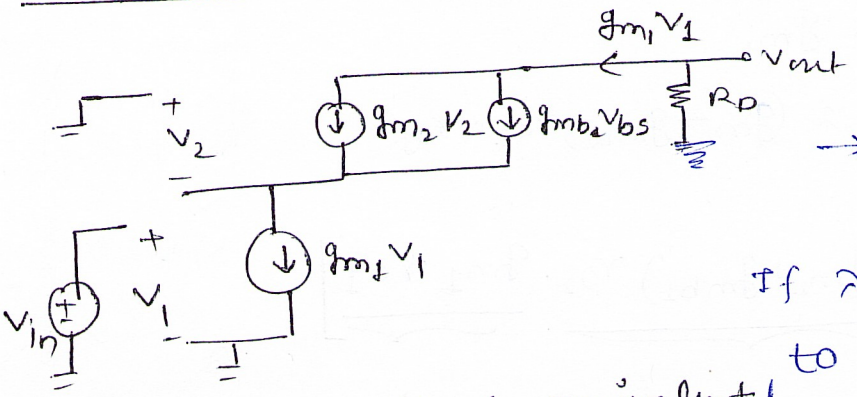
As V_{in} exceeds V_{TH1}, M_1 begin to draw current, V_{out} drop, I_{D_2} increases, V_{GS_2} must increases as well, causing V_x to fall.

As V_{in} assumes sufficiently large value two effect occure -

1) V_x drops below V_{in} by V_{TH_1}, forcing M_1 into the triode region.

② V_{out} drops below V_b by V_{TH_2}, driving M_2 into the triode region. Depending on the device dimensions and the values of R_D and V_b, one effect may occur before the other.

Small signal char. of a Cascode



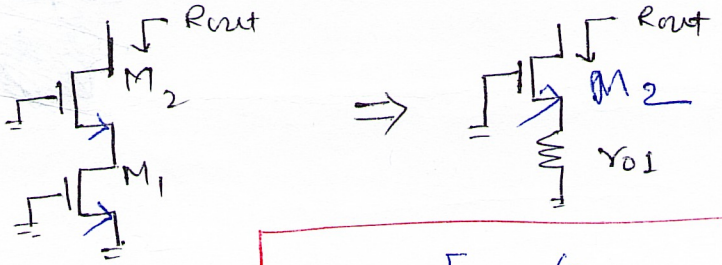
→ assume both transister operate in saturation.

Fig: Small signal equivalent circuit of cascode.

If $\lambda = 0$, the vtg gain is equal to that of a common source stage because the drain current produce by the input device must flow through the cascode device.

This results is independent of the transconductance & body effect of M_2

Imp. property of the cascode, is its high output impedence.



* Calculation of o/p resistance of cascode stage.

$$R_{out} = \left[1 + (g_{m_2} + g_{mb_2}) r_{o_2}\right] r_{o_1} + r_{o_2}$$

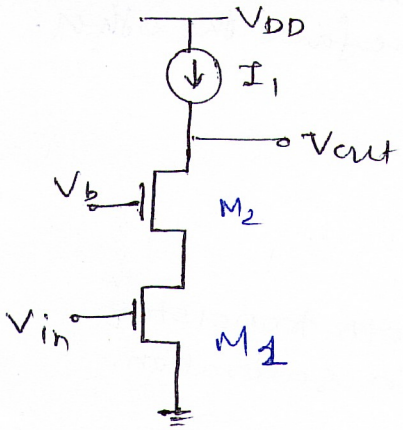
** calculation is same as eqn no. (20)

assume, $g_m r_o \gg 1$

$$\Rightarrow \quad R_{out} \simeq (g_{m_2} + g_{mb_2}) r_{o_2} * r_{o_1}$$

$\Rightarrow$ M_2 boosts the o/p impedence of M_1 by a factor of $(g_{m_2} + g_{mb_2}) r_{o_2}$.

Cascode Stage with current source load :-



→ Voltage gain can be written as

$G_M R_{out}$

$G_M \cong g_{m_1}$

$R_{out} \cong (g_{m_2} + g_{mb_2}) r_{O_2} \cdot r_{O_1}$

$\therefore \quad \boxed{A_V \cong (g_{m_2} + g_{mb_2}) r_{O_2} \cdot g_{m_1} r_{O_1}}$

⇒ The maximum voltage gain is roughly equal to the square of the intrinsic gain of the transistor.

Folded Cascode: